Blok 1 — Status & Fault Registers (0x0000–0x0028)

Toegang: Read-Only **Bron:** `lib/modbus/adlar-modbus-registers.ts` **Type:** 16-bit registers, grotendeels bitmaskers

Elk register is een 16-bit waarde. Bij de bitmasker-registers is elke bit een afzonderlijke status- of foutindicator. Een gezette bit (1) betekent dat de betreffende conditie actief is.

Registeroverzicht

Adres	Register	Type	Beschrijving
0x0000	Running Status 1	Bitmask	Bedrijfsstatus unit
0x0001	Running Status 2	Bitmask	Sterilisatie / controller status
0x0002	Fault State 1	Bitmask	Systeem- en sensorstoringen
0x0003	Fault State 2	Bitmask	Beveiligingen en pompstoringen
0x0004	Fault State 3	Bitmask	Communicatie- en sensorstoringen
0x0005	Sys 1 Fault State 1	Bitmask	Compressor/druk-storingen systeem 1
0x0006	Sys 1 Fault State 2	Bitmask	Druksensor-storingen systeem 1
0x0007	Sys 1 Drive Fault 1	Bitmask	Drive/omvormer-storingen systeem 1
0x0008	Sys 1 Drive Fault 2	Bitmask	Drive-storingen systeem 1 (v2.2)
0x0009	Sys 1 Drive Fault 3	Bitmask	Drive-storingen systeem 1 (v2.2)
0x000A–0x0018	Sys 2–4 Faults	Bitmask	Identieke structuur, +5 offset per systeem
0x0019	Relay Output Status 1	Bitmask	Relaisuitgangen (verwarming, ventilator, pompen)
0x001A	Relay Output Status 2	Bitmask	Relaisuitgangen (compressors, kleppen)
0x001B	Relay Output Status 3	Bitmask	Relaisuitgangen (v2.2)
0x001C	Relay Output Status 4	Bitmask	Relaisuitgangen pijp/pomp (v2.2)
0x001D	Switch Port State 1	Bitmask	Schakelaaringangen
0x001E	Switch Port State 2	Bitmask	Drukschakelaars (v2.2)
0x001F	Switch Port State 3	Bitmask	Buffervat AHS-koppeling (v2.2)
0x0027	Compressor Target Frequency 1	Waarde (Hz)	Doelfrequentie compressor 1
0x0028	Compressor Target Frequency 2	Waarde (Hz)	Doelfrequentie compressor 2

0x0000 — Running Status 1

Bit	Masker	Naam	Beschrijving
0	0x0001	REFRIGERANT_RECOVERY	Koudemiddel terugwinning
1	0x0002	PRIMARY_ANTIFREEZE	Primaire antivries
2	0x0004	SECONDARY_ANTIFREEZE	Secundaire antivries
3	0x0008	FAULT_ALARM	Storingsalarm
4	0x0010	SYSTEM_OIL_RETURN	Systeem olie terugvoer
8	0x0100	SYSTEM_DEFROST	Ontdooiing actief
12	0x1000	CONST_TEMP_SHUTDOWN	Constante temp bereikt, compressor uit
13	0x2000	FAULT_SHUTDOWN	Uitgeschakeld na storing
14	0x4000	MACHINE_RUN	Unit in bedrijf
15	0x8000	MACHINE_WAIT	Unit wacht op operatie

0x0001 — Running Status 2

Bit	Masker	Naam	Beschrijving
0	0x0001	HIGH_TEMP_STERILIZATION	Legionella cyclus actief
1	0x0002	HIGH_TEMP_STERIL_PRESERVE	Sterilisatie warmhouden
10	0x0400	CONTROLLER_ON_OFF	Controller aan/uit

0x0002 — Fault State 1

Bit	Masker	Naam	Beschrijving
0	0x0001	WRONG_PHASE	Verkeerde fase
1	0x0002	LACK_OF_PHASE	Fase ontbreekt
2	0x0004	WATER_FLOW	Waterstroom fout
3	0x0008	COMMUNICATION	Communicatie fout
4	0x0010	EMERGENCY	Noodstop
5	0x0020	USE_TIME_EXPIRED	Gebruikstijd verlopen
6	0x0040	WATER_TANK_TEMP	Boiler temp sensor fout
7	0x0080	WATER_INLET_TEMP	Inlaat temp sensor fout
8	0x0100	INDOOR_TEMP	Binnentemp sensor fout
9	0x0200	ENVIRONMENTAL_TEMP	Buitentemp sensor fout
10	0x0400	USER_BACKWATER_TEMP	Retour temp sensor fout
11	0x0800	COOLING_OUTLET_LOW	Koeluitlaat te laag
12	0x1000	WATER_LEVEL_SWITCH	Waterstand schakelaar fout
13	0x2000	WATER_OUTLET_TEMP	Uitlaat temp sensor fout
14	0x4000	HEATING_OUTLET_HIGH	Verwarmingsuitlaat te hoog
15	0x8000	EXCESSIVE_TEMP_DIFF	Te groot temp verschil in/uit

0x0003 — Fault State 2

Bit	Masker	Naam	Beschrijving
0	0x0001	ENV_LOW_TEMP_PROTECT	Lage buitentemp beveiliging
6	0x0040	INDOOR_HUMIDITY	Luchtvochtigheid sensor fout
11	0x0800	PHASE_ORDER_DIP	Fase-volgorde DIP fout
13	0x2000	WATER_PUMP_1_FEEDBACK	Waterpomp 1 terugkoppeling fout
14	0x4000	WATER_PUMP_2_FEEDBACK	Waterpomp 2 terugkoppeling fout
15	0x8000	LOW_WATER_FLOW	Te lage waterstroom

0x0004 — Fault State 3

Bit	Masker	Naam	Beschrijving
0	0x0001	PHASE_SEQ_DISCONNECT	Fase-volgorde verbroken
1	0x0002	EXPANSION_BOARD_COMM	Uitbreidingsboard communicatie
2	0x0004	PLATE_HX_TEMP	Platenwisselaar temp fout
3	0x0008	FAN_MOTOR_1_COMM	Ventilatormotor 1 communicatie
4	0x0010	FAN_MOTOR_2_COMM	Ventilatormotor 2 communicatie
5	0x0020	ONLINE_MODEL_MISMATCH	Model niet overeenkomend
6	0x0040	SOLAR_HW_SENSOR	Solar warm water sensor fout *(v2.2: was "AUX_HW_SENSOR")*
7	0x0080	AHS_TEMP_SENSOR	AHS temp sensor fout *(v2.2: naam verduidelijkt)*
8	0x0100	BUFFER_TANK	Buffervat temp sensor fout

Bit	Masker	Naam	Beschrijving
9	0x0200	MAIN_OUTLET_TEMP	Hoofd uitlaat temp fout
12	0x1000	ZONE_1_TEMP_SENSOR	Zone 1 temp sensor fout *(v2.2)*

Systeem 1 Compressor/Drive Faults (0x0005–0x0009)

Systemen 2–4 hebben dezelfde structuur op adressen 0x000A–0x0018 (+5 offset per systeem).

0x0005 — Sys 1 Fault State 1

Bit	Masker	Naam	Opmerking
0	0x0001	HIGH_PRESSURE_SWITCH	
1	0x0002	LOW_PRESSURE_SWITCH	
2	0x0004	HIGH_PRESSURE_OVER	
3	0x0008	LOW_PRESSURE_OVER	R290: “High Pressure Too Low” = drukval
4	0x0010	EXHAUST_OVER	
5	0x0020	CURRENT_PROTECTION	
6	0x0040	COIL_PRESSURE_HIGH	v2.2: was “COIL_TEMP_OVER”; R290 “Coil Pressure Too High”
7	0x0080	COIL_TEMP_FAULT	
8	0x0100	RETURN_AIR_TEMP_FAULT	Suction temp sensor
9	0x0200	EXHAUST_TEMP_FAULT	
10	0x0400	ECONOMIZER_INLET_FAULT	
11	0x0800	ECONOMIZER_OUTLET_FAULT	
12	0x1000	FAN_DRIVE_COMM	
13	0x2000	DC_FAN_FAULT	
14	0x4000	REFRIG_COIL_TEMP_FAULT	Cooling coil temp sensor

0x0006 — Sys 1 Fault State 2

Bit	Masker	Naam
0	0x0001	HIGH_PRESSURE_SENSOR
1	0x0002	LOW_PRESSURE_SENSOR
2	0x0004	MIDDLE_PRESSURE_SWITCH
3	0x0008	COIL_TEMP_OVER_HIGH
4	0x0010	COMP_DRIVE_BOARD_COMM

0x0007 — Sys 1 Drive Fault 1

Bit	Masker	Naam	Opmerking
0	0x0001	IPM_OVERCURRENT	
1	0x0002	COMPRESSOR_DRIVE	
2	0x0004	COMPRESSOR_OVERCURRENT	
3	0x0008	INPUT_VOLTAGE_LOSS	
4	0x0010	IPM_CURRENT_SAMPLING	
5	0x0020	POWER_COMP_OVERHEAT	
6	0x0040	PRECHARGE_FAILED	
7	0x0080	DC_BUS_OVERVOLTAGE	

Bit	Masker	Naam	Opmerking
8	0x0100	DC_BUS_UNDERVOLTAGE	
9	0x0200	AC_INPUT_UNDERVOLTAGE	
10	0x0400	AC_INPUT_OVERVOLTAGE	v2.2: was “OVERCURRENT”; R290 “Overvoltage”
11	0x0800	INPUT_VOLT_SAMPLING	
12	0x1000	DSP_PFC_COMM	
13	0x2000	RADIATOR_TEMP_SENSOR	
14	0x4000	DSP_COMM_BOARD	
15	0x8000	MAIN_CONTROL_BOARD	

0x0008 — Sys 1 Drive Fault 2 ***(v2.2)***

Bit	Masker	Naam
0	0x0001	COMPRESSOR_OVERCURRENT_ALARM
1	0x0002	WEAK_MAGNETIC_PROTECTION
2	0x0004	PIM_OVERHEAT
3	0x0008	PFC_OVERHEAT
4	0x0010	AC_INPUT_OVERCURRENT
5	0x0020	EEPROM_ERROR
7	0x0080	EEPROM_REFRESH_COMPLETE
8	0x0100	TEMP_SENSING_LIMIT
9	0x0200	AC_UNDERVOLT_FREQ_LIMIT

0x0009 — Sys 1 Drive Fault 3 ***(v2.2)***

Bit	Masker	Naam
0	0x0001	IPM_OVERHEAT_SHUTDOWN
1	0x0002	COMPRESSOR_MISSING_PHASE
2	0x0004	COMPRESSOR_OVERLOAD
3	0x0008	INPUT_CURRENT_SAMPLING
4	0x0010	PIM_SUPPLY_VOLTAGE
5	0x0020	PRECHARGE_VOLTAGE
6	0x0040	EEPROM_FAILURE
7	0x0080	AC_INPUT_OVERVOLTAGE
8	0x0100	MICROELECTRONICS
9	0x0200	COMPRESSOR_TYPE_CODE
10	0x0400	CURRENT_SAMPLING_OVERCURRENT

Relaisuitgangen (0x0019–0x001C)

0x0019 — Relay Output Status 1

Bit	Masker	Naam	Beschrijving
0	0x0001	HOT_WATER_EHEATING	Warm water elektrische bijverwarming
1	0x0002	FAN_HIGH_WIND	Ventilator hoog toerental
3	0x0008	FAN_LOW_WIND	Ventilator laag toerental
4	0x0010	AC_EHEATING	Airco elektrische bijverwarming
5	0x0020	FLOOR_EHEATING	Vloer elektrische bijverwarming

Bit	Masker	Naam	Beschrijving
6	0x0040	MAIN_CIRCULATING_PUMP	Hoofdcirculatiepomp
9	0x0200	ELEC_CRANKSHAFT_HEATING	Krukas verwarming
10	0x0400	CHASSIS_EHEATING	Chassis verwarming
11	0x0800	RETURN_VALVE_PUMP	Retourklep pomp
14	0x4000	AC_SOLENOID_3WAY	Airco 3-weg magneetklep
15	0x8000	FLOOR_SOLENOID_3WAY	Vloer 3-weg magneetklep

0x001A — Relay Output Status 2

Bit	Masker	Naam	Opmerking
0	0x0001	COMPRESSOR_1	
1	0x0002	LIQUID_INJECTION_1	
2	0x0004	EVI_EEV_1	v2.2: was “ENTHALPY_SOLENOID_1”
3	0x0008	FOUR_WAY_VALVE_1	
4	0x0010	BYPASS_VALVE_1	v2.2: was “THROTTLE_BYPASS_1”
5	0x0020	FAN_MOTOR_1	
8	0x0100	AUX_HEATING_PUMP	R290: “Secondary heating pumps”
10	0x0400	COMPRESSOR_2	
11	0x0800	LIQUID_INJECTION_2	
12	0x1000	EVI_EEV_2	
13	0x2000	FOUR_WAY_VALVE_2	v2.2: was op 0x4000

0x001B — Relay Output Status 3 *(v2.2)*

Bit	Masker	Naam
6	0x0040	EXPANSION_TANK_EHEATING
7	0x0080	HW_HEAT_SOURCE_PUMP
8	0x0100	HEATING_HEAT_SOURCE_PUMP
9	0x0200	AHS_SIGNAL_OUTPUT

0x001C — Relay Output Status 4 *(v2.2)*

Bit	Masker	Naam
0	0x0001	PIPE_EHEATING_1
1	0x0002	PIPE_EHEATING_2
2	0x0004	AUX_WATER_PUMP
3	0x0008	ZONE_2_WATER_PUMP
4	0x0010	ZONE_1_WATER_PUMP

Schakelaaringangen (0x001D–0x001F)

0x001D — Switch Port State 1

Bit	Masker	Naam	Beschrijving
0	0x0001	SW1	
1	0x0002	SW2	
2	0x0004	SW3	

Bit	Masker	Naam	Beschrijving
3	0x0008	SW4	
4	0x0010	SW5	
5	0x0020	SW6	
6	0x0040	SW7	
7	0x0080	SW8	
8	0x0100	WATER_FLOW_SWITCH	Waterstroomschakelaar
10	0x0400	HOUSE_HEATING_LINKAGE	Kamerthermostaat
11	0x0800	AUX_HW_LINKAGE	DHW AHS
12	0x1000	LINKAGE_SWITCH	Koppelschakelaar
13	0x2000	EMERGENCY_SWITCH	Noodschakelaar

0x001E — Switch Port State 2 *(v2.2)*

Bit	Masker	Naam
7	0x0080	HIGH_PRESSURE_SWITCH_1
8	0x0100	LOW_PRESSURE_SWITCH_1
9	0x0200	MIDDLE_PRESSURE_SWITCH_1
10	0x0400	HIGH_PRESSURE_SWITCH_2
11	0x0800	LOW_PRESSURE_SWITCH_2
12	0x1000	MIDDLE_PRESSURE_SWITCH_2

0x001F — Switch Port State 3 *(v2.2)*

Bit	Masker	Naam
5	0x0020	BUFFER_TANK_AHS_LINKAGE

Compressor doelfrequenties

Adres	Naam	Eenheid
0x0027	Compressor target frequency 1	Hz
0x0028	Compressor target frequency 2	Hz